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Fair Evaluator: An Adversarial Debiasing-based Deep Learning Framework in Student Admissions

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***Abstract*—This work considers the problem of enhancing the authenticity and fairness of undergraduate student admission decision-making process by employing state-of-the-art Deep Learning (DL) models with advanced bias-mitigation techniques. Traditional admission processes often introduce biases that can disadvantage underrepresented or marginalized groups, highlighting the need for more equitable and efficient methods. Although the DL models have emerged as a promising alternative offering superior performance and higher scalability than the classical Machine Learning approaches, fairness concerns remain a significant issue.**

We propose an adversarial debiasing-based DL framework that integrates the Optimistic Adam (OAdam) optimizer, ensuring consistent and stable model training crucial for achieving reliable and unbiased outcomes. Our framework leverages data from applicants to the Computer Science Department at the University of California, Irvine. To ensure holistic evaluation of applicants' profile we utilize a dataset that encompasses a wide range of features showcasing demographics, academic records, high school information, and essay responses. By prioritizing the recall score alongside the fairness metrics, our approach effectively handles the fairness-accuracy trade-off, considerably minimizing the false negatives and ensuring equitable consideration for marginalized groups in admission decisions. Through rigorous experimentation and analysis, our comprehensive study demonstrates that the proposed fairness-aware Input Convex Neural Network model using OAdam optimizer, achieves high fairness metrics while ensuring a balanced predictive performance. The proposed model improves the *p-%* rule scores by an average of 39.989% across sensitive attributes and achieves recall scores 0.97% higher than those of unfair baseline models.

***Index Terms*—fairness, deep learning, adversarial debiasing, undergraduate student admissions, holistic evaluation**

I. Introduction

Over the past decades, the undergraduate student admission processes have undergone notable transformations, with a pronounced emphasis on fostering fairness and equal opportunity for all applicants. These changes are particularly evident within the educational frameworks of the University of California (UC). Certain initiatives such as the elimination of race and gender considerations, implementation of percentage plans to recognize top-performing students, and removal of standardized testing, showcase the effort of the system to prioritize inclusivity [1]. While these measures represent strides toward a more equitable system, there still exists a lot of challenges pertaining to the socioeconomically disadvantaged groups. The disparity in resources and support available to these groups often results in unequal academic performance, further aggravating the existing inequalities in the admission processes. To further our case, [2] and [3] explain the systematic problems related to student admissions and the various other factors beyond academic merit, such as the socioeconomic factors, that may adversely impact the decision-making process. Furthermore, [4] describes the unfair decision-making process due to the historical bias present in the data. Recognizing the importance of addressing these systemic barriers, our study seeks to investigate and integrate fairness in the decision-making process, aiming to provide insightful solutions in the domain of undergraduate student admissions.

The incorporation of Machine Learning (ML) techniques in the undergraduate admissions decision-making process is aimed towards the removal of any human bias and prejudice. However, these models are still limited by their inability to capture complex patterns within high-dimensional data. Unlike the classical ML models, the Deep Learning (DL) models excel at handling high-dimensional data while also capturing the intricate patterns within them. The DL models are well suited for processing and extracting meaningful information from the applicants' data for our study, as asserted by [5]. In the context of undergraduate student admissions, the prediction of admission outcomes relies on diverse and multifaceted applicant profiles. Therefore, the efficiency of the DL models provides significant advantages for this task.

By integrating DL models with fairness techniques, as outlined in [6], we not only ensure fairness and equity in the admissions decision-making process but also optimize the DL models to maintain their predictive performance. For our study, we have incorporated the adversarial debiasing techniques [7] to optimize the DL models to adhere to the fairness constraint. Adversarial debiasing is a type of fair representation learning that aims at mitigating bias by simultaneously minimizing the predictive errors and maximizing the discrepancies between the prediction and the sensitive attributes present in the dataset. Overall, the technique offers an efficient approach to address the bias present in the applicants' data, while streamlining the decision-making process for improved performance.

In this work, we have analyzed 4,442 application records

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of California freshman applicants for the Fall, 2021 cycle to the Department of Computer Science at the University of California, Irvine (UCI). The dataset comprises various variables including demographics, academic records, high school information, and essay question responses. Additionally, we acknowledge the limitations imposed due to the restricted amount of training data for the study. The usage of admissions data from the Fall 2021 cycle is attributed to the sensitive nature of the admissions data that contains personal information of applicants to the UCI. As a result, acquiring such data is particularly challenging. Given these constraints, we aim to explore the potential of ML models coupled with fairness constraints to assist with holistic undergraduate admissions. In this context, *holistic evaluation* signifies a comprehensive approach to student admissions that assesses applicants based on a wide range of criteria beyond academic performance. This consideration of a broader spectrum of criteria in holistic evaluation helps provide a more equitable and fairer assessment [8]. To ensure this, the training data includes diverse features providing a comprehensive view of the applicant's profile. As articulated in the policy [1], factors such as race, color, national origin, religion, sex, gender, pregnancy status, disability, or medical condition do not impact the decision-making procedure. However, it is equally crucial to acknowledge the socioeconomic status of the student which has not been explicitly mentioned in the policy. As a result, models trained on such data may inadvertently perpetuate these biases, leading to unfair and discriminatory outcomes. In this study, we try to handle such issues by developing novel methods and strategies aimed at reducing the influence of any socioeconomic and related representation bias present in the admissions data for our DL-based decision-making process. Our key contributions are as follows:

- We propose a systematic analysis of features in our input data to identify the ones that reflect the applicant's socioeconomic status. Subsequently, we evaluate the presence of bias using fairness metrics and Kernel Density Estimation plots. The features propagating socioeconomic bias are chosen as the sensitive attributes for the study.
- We propose the incorporation of adversarial debiasing, an in-processing technique (characterized based on intervention phase of the debiasing algorithm), in DL models for the admissions decision-making process. The fairness-aware DL models promise to enhance the fairness quotient while maintaining the predictive performance.
- We try to address the inherent fairness-accuracy trade-off by employing the Optimistic Adam optimizer. This ensures a balanced fair predictive performance of the DL models in a zero-sum game setting.

II. Related Works

In recent years, the pursuit of fairness in ML models has been gaining significant attention, particularly in the areas where the decision has profound implications on societal demographics [9]. Fairness-aware DL has emerged as a critical area of research, particularly in the context of student admissions, where the goal is to mitigate bias and promote equity.

A. Fairness in Student Admissions

Several studies have highlighted various aspects of fairness in admissions. For example, [10] emphasizes the complexities in quantifying fairness and the need for robust evaluation criteria for fairness in the decision-making processes. On the other hand, there exists a lot of work focusing on surveying real-world scenarios in student admissions decision-making. [11] explores the student selection processes in the context of fairness and highlights the importance of considering diverse perspectives in fairness assessments. Similarly, [3] and [12] talk about the consequences of perceived fairness or unfairness of artificial intelligence (AI) systems in admissions but do not delve into the underlying mechanisms or potential biases embedded within the AI system or provide any solution. On similar lines of study, [13] offers a psychometric examination of bias and fairness but does not describe other complex socio-economic factors affecting student admissions. Although [2] discusses the various factors beyond academic merit, influencing the admissions decisions, their reliance on traditional statistical methods makes it less rigorous. Qualitative explorations conducted in [14] offer valuable insights into the subjective nature of fairness perceptions and decision-making processes in undergraduate admissions but are unable to capture the quantitative metrics used in modern decision-making processes. Similarly, [15] addresses the admissions problem being related to the historical biases present in the data but falls short of providing comprehensive methodologies to address such biases effectively. Although the mentioned studies provide insights into the broader issues of fairness in student admissions, there exists notable lack of robust, comprehensive approaches specifically designed to tackle nuanced fairness challenges inherent in this domain.

B. Applications of Machine Learning in Student Admissions

Despite these contributions, there remains a significant gap in robust methodologies to assess and mitigate biases adequately. Many existing approaches using DL and off-the-shelf ML approaches, like [16], [17], [18], and [19], overlook the fairness considerations, leading to incomplete or biased outcomes. Additionally, the static nature of fairness metrics used in some studies may not fully capture the evolving dynamics of decision-making processes in a real-world setting [20]. Furthermore, the focus on academic merit and standardized criteria in admission decisions may inadvertently incorporate systemic biases and inequalities, especially affecting marginalized or underrepresented groups.

Our study aims to fill the existing gaps in student admissions literature by leveraging advanced ML techniques to create a more equitable admissions process. Following the work established in [5], which leverages a DL-based model for admission prediction in undergraduate applications, we propose to extend this framework by integrating adversarial debiasing techniques. This implementation represents a novel approach

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in the domain of student admissions, given that existing methods primarily overlook the underlying mechanisms of bias in learning process and fail to provide robust solutions to mitigate them. By addressing the limitations identified in prior studies, our approach offers a comprehensive solution to the fairness challenges inherent in student admissions decision-making processes.

III. Preliminaries

A. Fairness in Machine Learning

ML algorithms have become an integral tool across diverse domains. As these models are being increasingly used for informed decision-making processes, concerns regarding fairness and equity have gained significant attention. In the context of decision-making, fairness can be described as the absence of any prejudice towards an individual or a group based on their intrinsic or acquired traits [6]. Fair ML models strive to avoid any influence, either advantageous or disadvantageous, of any demographic group based on protected attributes like sex, gender, or any other socioeconomic features. For our study, this entails ensuring equitable treatment and outcomes, mitigating algorithmic biases, and promoting transparency and accountability of the model, while upholding ethical principles in the algorithmic decision-making process [21]. To formalize and quantify fairness within our framework, we employ two widely recognized fairness metrics namely demographic parity and $p\%$-rule. A comprehensive discussion of these metrics, including their definitions and relevance to our approach, is provided in Section IV-C.

B. Why do we need Fairness in Admissions?

The paper [6] discusses the various sources of bias in ML models and categorizes them into three types namely, data-to-algorithms, algorithm-to-user, and user-to-data. In the study for student admissions decision-making, we are dealing with data-to-algorithm type bias. Specifically put, our case study is affected by two types of biases namely, *socioeconomic bias* and its related *representation bias*. Socioeconomic bias might arise due to the disproportionate influence of input features representing the applicant's socioeconomic status. On the other hand, representation bias might arise during the sampling of data where there exists a high chance of collecting non-representative samples that lack the diversity of the population. Apart from that, measurement bias could be induced by individuals from minority communities due to their past/family records. This might affect their University acceptance rate. In regards to the user-to-data type bias, historical bias also has a significant impact on our case study.

Apart from bias, discrimination is also considered a significant source of unfairness. This emerges due to human prejudice and stereotyping based on sensitive attributes present in the data, which may or may not be intentional. Discrimination can be categorized into two classes namely, disparate treatment and disparate impact, as mentioned in [6], [22]. Our study falls under the category of *disparate impact* (DI) which refers to a situation where the policy or the decision-making process has an adverse impact on the members of a protected group or marginalized community, even if there is no such intent. Mathematically, DI could be visualized as,

$$DI = \frac{P(\text{outcome} \mid \text{protected group=1})}{P(\text{outcome} \mid \text{protected group=0})}$$

In summary, both bias and discrimination can lead to perpetuating systematic inequalities, further undermining trust in ML models. By addressing fairness concerns, we try to promote equitable student admissions decision-making while leveraging the ML model's potential.

C. Categories of Fairness

Bias mitigation can be handled using various methodologies proposed. Depending on the intervention phase of the debiasing algorithm we can categorize it into three classes, namely *pre-processing*, *in-processing*, and *post-processing* modeling techniques. Our study focuses on the in-processing technique for bias mitigation. The rationale behind the choice has been outlined as follows [23],

1) The in-processing technique addresses the problem of bias amplification in model training. It takes fairness directly into consideration for model optimization so that the converged model can achieve fairness even with biased data as input.
2) This technique focuses on addressing fairness issues under identifiable sensitive attributes present in the input data.

For our study, we consider an *implicit in-processing technique* that focuses on debiasing so that the resulting predictions do not show discrimination towards the under-represented groups. By leveraging adversarial learning in the in-processing method, we enforce a fairness-oriented adversary along with the classifier model. The proposed approach can be conceptualized as a *min-max* optimization problem, aiming to maximize the classifier's ability to predict the labels while simultaneously minimizing the adversary's ability to predict the sensitive attributes from the former's predictions. By harnessing the optimization techniques, the in-processing method offers a systematic approach to attain fairness in ML models.

IV. Methodology

In this section, we outline the methodology employed in our study, which involves discussing the architectures of two types of deep neural networks (DNN), the application of adversarial debiasing techniques, and the Optimistic Adam optimizer. Our approach leverages adversarial debiasing technique along with the DL models to mitigate bias in the prediction outcomes, thereby contributing to the advancement of an equitable admission process. Additionally, we utilize the Optimistic Adam (OAdam) optimizer to enhance the training process.

A. Data preprocessing

In this study, we present a comprehensive analysis of DL models using in-processing-based fair representation learning to mitigate biases on a dataset of 4,442 applications to the

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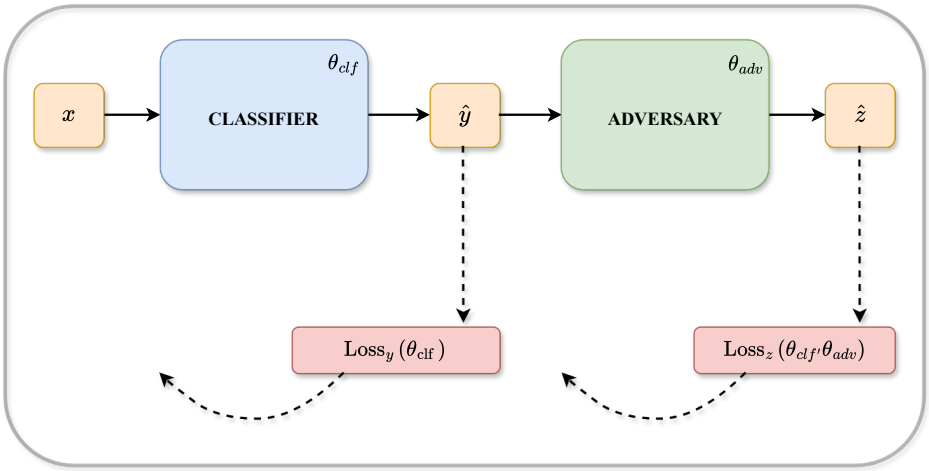


Figure 1: The generalized architecture of the Adversarial Debiasing Network. The Classifier model is selected as either a FFNN, or an ICNN, and the Adversary model is chosen as a FFNN.

Dept. of Computer Science at the UCI. The dataset [19] contains various categories of information, including the student Grade Point Average (GPA), Advanced Placement (AP) test scores, participation in educational programs, and various responses to the admission-related questions. We use the final read score as the target variable for our binary classification problem, which represents the review score assigned to every applicant. For our study, the top review score is mapped to 1, and the lower scores are mapped to 0. With this approach, we seek to identify the applications that are likely to receive high scores provided we ensure the dataset is balanced with about 50% of records belonging to both classes.

Entries missing the final read score values or other numerical data, like the GPA scores, were excluded from the study to ensure model training and performance evaluation efficacy. To enhance the efficiency of the model training process, records of students with high school attendance outside California (CA) were dropped. The motivation for this approach is that the CA applicants' records includes a variety of potential information regarding the high school of attendance as well as Eligibility in the Local Context (ELC) program, that recognizes the accomplishments of top 9% of the students. It is of utmost importance to include these features as it helps ensure that our goals align with the holistic review standards. Next, we handle any missing entries through median imputation and further normalize to generate a well-articulated training dataset. Additionally, we employ data scaling that maps the normalized data to a range enhancing compatibility with algorithms, such as the DNN models, reliant on appropriate input ranges. Through our work we try to ensure that the proposed DL model upholds the data integrity during its training and evaluation phases.

However, it is important to note that the data still consists of features, that inherently capture aspects such as family income levels, parental education, and other indicators of socioeconomic backgrounds, that may impact minority groups, particularly those with socioeconomic disadvantages or historical disparities. We avoid dropping features representing the applicant's socioeconomic status to ensure overall profile consideration in light of holistic evaluation. Also, dropping a set of sensitive features from the input data does not guarantee fairness in prediction [7], [24], as the dropped features might have an indirect influence on the existing features eventually leading to bias induction in the learning model.

B. Neural Network Architecture

In this study we employ two distinct neural network architectures namely, Feed-forward Neural Network (FFNN) and Input Convex Neural Network (ICNN) for our classifier model [5].

1) Feed-forward Neural Network: The FFNN is a fundamental architecture characterized by a unidirectional flow of information [25]. Owing to its network architecture, it is capable of efficiently modeling complex relationships in the data and making predictions that are easy to interpret. For our study, we incorporate a four-layered FFNN, as the classifier model, for the adversarial debiasing technique to mitigate bias present in the data. The scalability feature along with the model's high interpretability makes it well-suited for the problem.

Definition 1 (Feed-Forward Predicted Output)**.** The predicted output of the network, p, is defined as follows,

$$p = \sigma[W_k \cdot f_k(W_{k-1} \cdot f_{k-1}(...f_1(W_1 \cdot x + b_1) + ...) + b_k]$$

where σ is the sigmoid function on the outermost layer of the FFNN, and for all $1 \leq i \leq k$, W_i are the weight vectors, b_i are the bias vectors, and f_i are the activation functions respectively, provided k is the number of hidden layers in the FFNN.

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2) Input Convex Neural Network: The ICNN is a scalar-valued neural network with constraints on its parameters, ensuring that the output of the network is a convex function of the input [26]. The architecture ensures that the network is capable of explicitly capturing the convex relationships between the input variables, thereby preserving the convexity property. Because of the network's capability to preserve complex relationships, including convex ones, ICNN facilitates the detection and mitigation of biases that might be present in the data. In this work, we consider a fully connected ICNN (FICNN) model.

Definition 2 (FICNN Predicted Output)**.** Say, $f(x;\theta)$ is the scalar-valued neural network where x denotes the input to the function and θ are the parameters such that the network, f, is convex w.r.t the input x. In order to understand the predictions, the layer-wise output is calculated as,

$$z_{i+1} = g_i(U_i.z_i + W_i.x + b_i)$$

where W_i are the real-valued weights mapping from inputs to the $i+1$ layer activations; U_i are positive weights mapping previous layer activations z_i to the next layer; b_i are the real-valued bias terms; and g_is are convex, monotonically non-decreasing non-linear activation functions for every i representing the training samples. Then the predicted output of the network, p, could be defined as follows,

$$p = g(x) \equiv z_k$$

where k signifies the number of layers in the network.

We also implement the dropout regularization to enhance its generalization capabilities and mitigate the risk of overfitting. This technique aims to randomly drop fractions of neurons with a probability of $d \in [0, 1]$. This technique introduces randomness into the training process that helps the model learn more diverse and representative features, which are less influenced by any bias present in the data. This, in turn, leads to fairer predictions on unseen test data.

C. Adversarial Debiasing

In our framework, we consider an adversarial debiasing technique [7] that takes the supervised DL task and implements an adversary along with the classifier to handle the algorithmic bias induced into the model by the data. The task at hand involves predicting an output variable Y given an input variable X while remaining unbiased with respect to the variable Z. Here, Z is referred to as protected attributes, that affect the decision-making process by inducing bias into the model. Figure 1 illustrates the architecture of the adversarial network and its mechanism. For our binary classification problem, we consider a classifier model that makes predictions $\hat{y}$ with input data X, and also an adversary model tasked with predicting whether the classifier exhibits unfair behavior with respect to the sensitive attributes, Z. As discussed previously, we employ two distinct DNNs, namely FFNN and ICNN, respectively as the classifier model, and for the adversary model we deploy the FFNN, provided it is a simple and interpretable architecture.

The classifier engages in a zero-sum game with the adversary, aiming to achieve accurate predictions while trying to avoid penalization if the adversary detects unfair decisions. The goal of this technique is to develop a fair classifier capable of maintaining a predictive performance while being scrutinized by the adversary, aimed at detecting unfairness.

Definition 3 (Zero-Sum Game)**.** The loss function of the adversarial debiasing technique, following a zero-sum game approach, could be defined as,

$$\min_{\theta_{clf}}\{\text{Loss}(\theta_{clf}) - \lambda \cdot \text{Loss}(\theta_{clf}, \theta_{adv})\}$$

where $\lambda > 0$ is the hyperparameter controlling the strength of the fairness constraint, θ_{clf} and θ_{adv} represent the parameters of the classifier and the adversary respectively.

Following the definition, the first term $\text{Loss}(\theta_{clf})$ represents the classifier's standard loss function that aims at minimizing the prediction error. Meanwhile, the second term $\text{Loss}(\theta_{clf}, \theta_{adv})$ captures the adversarial component, where the classifier's predictions are penalized if the adversary detects unfairness. So the objective of the learning algorithm is to maximize its ability to predict Y while simultaneously minimizing its sensitivity around Z. Next, the fairness constraint λ acts as a crucial factor in balancing predictive accuracy and fairness. By adjusting λ, the trade-off between maximizing predictive accuracy and mitigating bias can be fine-tuned to fit the model while enabling flexibility in model optimization [27]. The fairness parameter ensures that the resulting model achieves the desired levels of fairness without significantly compromising the classifier's performance. Based on empirical evaluations, we set the value of λ to 50 for our experiments. The selection enforces balancing the trade-off between minimizing classification loss and effectively enforcing the fairness constraint.

In our analysis of the student admissions problem, we evaluate the fairness of our model using two prominent fairness metrics,

- For *Demographic parity* or statistical parity, a binary classifier $\hat{y}$ satisfies equal opportunity with respect to Z and y if the protected and the unprotected groups have equal true positive rates [6], [7]. Mathematically the following equation must satisfy for the principle to hold,

$$P(\hat{y} = 1|Z = 0, y = 1) = P(\hat{y} = 1|Z = 1, y = 1)$$

- For *p%-rule*, the ratio between the percentage of subjects having a certain sensitive attribute value assigned the positive decision outcome and the percentage of subjects not having that value also assigned the positive outcome should be no less than $p : 100$ [28]. Mathematically it must satisfy the following equation,

$$|P(\hat{y} = 1|Z = 0, y = 1) - P(\hat{y} = 1|Z = 1, y = 1)| \leq \epsilon$$

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Table I: Performance Metrics for Unfair Baseline Models

Learning Model	*Accuracy*	*Precision*	*Recall*	*F1-Score*	*AU-ROC Score*
Logistic Regression	0.783	0.762	0.802	0.781	0.871
XGBoost	0.7761	0.7711	0.7729	0.7720	0.7760
Feed-forward Neural Network	0.8056	0.8159	0.7883	0.8018	0.8056
Input Convex Neural Network	0.8067	0.8281	0.7961	0.8118	0.8073

Table II: Performance Metrics for Fair Deep Learning Models

Learning Model	*Accuracy*	*Precision*	*Recall*	*F1-Score*	*AU-ROC Score*
Using ADAM Optimizer					
Feed-forward Neural Network	0.7255	0.7028	0.7848	0.7415	0.7253
Input Convex Neural Network	0.7323	0.6912	0.8430	0.7596	0.7319
Using OADAM Optimizer					
Feed-forward Neural Network	**0.7525**	**0.7415**	0.7780	0.7593	**0.7524**
Input Convex Neural Network	0.7424	0.7140	**0.8117**	**0.7597**	0.7422

where $\epsilon \in [0, 1]$, $\hat{y}$ is the predicted outcome, y is the true label, and Z is the sensitive attribute. The $p\%$-rule is said to be satisfied when $\epsilon = p/100$.

The $p\%$-rule serves to bolster demographic parity by providing a quantitative measure of fairness that complements the qualitative assessment provided by the latter. While demographic parity promotes equal treatment among demographic groups, the $p\%$-rule ensures a balanced ratio of positive outcomes across different sensitive attribute values. This combined approach not only enhances the comprehensiveness of our fairness assessment but also emphasizes promoting transparency and accountability in the algorithmic decision-making process. In our study, we aim to maximize the model's predictive performance subject to fairness constraints, while enabling compliance with disparate impact doctrine in its basic form, *i.e.* the $p\%$-rule. These formulations of fairness avoid disparate impact as well as disparate treatment since they do not make use of sensitive attribute information in the decision-making process.

D. Optimistic Adam

For our model training, we extensively utilize the Adam optimizer [29] which is a first-order gradient-based optimization technique. Irrespective of its efficiency in handling non-convex optimizations and adaptive learning rate capabilities, we observed fluctuations in loss function impeding the overall training process. To circumvent this issue, we utilize the Optimistic Adam (OAdam) optimizer [30] for our case study resulting in comparatively stable convergence and efficient handling of parameter updation during the neural network training process. The OAdam introduces an optimistic update term, $\triangle\theta_t$, in the parameter update equation which aims to explore diverse modes in the data distribution and also lead to enhanced stability during training allowing the model to converge to fairer and unbiased solutions more reliably.

Definition 4 (Optimistic Adam weight update)**.** The OAdam algorithm [30] proposes a weight update technique that can be described as follows,

$$\theta_t = \theta_{t-1} - 2\eta \cdot s_t + \eta \cdot s_{t-1}$$

where η is the step size, θ_t and θ_{t-1} are the parameters of the DNN model at time step t and $t-1$ respectively. Similarly, s_t and s_{t-1} are the scaled gradients at time step t and $t-1$ respectively. The scaled gradients are the gradients of the loss function with respect to the model parameters, adjusted by the learning rate. This ensures that the parameter updates are well-scaled for stable and efficient convergence.

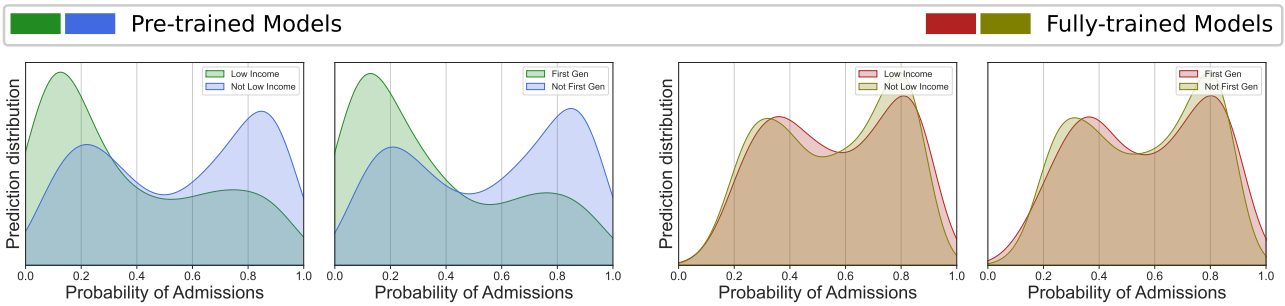
In the context of adversarial debiasing, OAdam offers various benefits such as enhanced stability, dynamic learning rate, robustness to adversarial attacks, and efficient optimization, hence providing improved resilience to biases. By stabilizing convergence behavior and preventing overfitting, OAdam helps train the model to generalize well to unseen data, including diverse and underrepresented samples. The efficiency of OAdam for fairness-aware ML models is further affirmed by [22].

V. Experiments

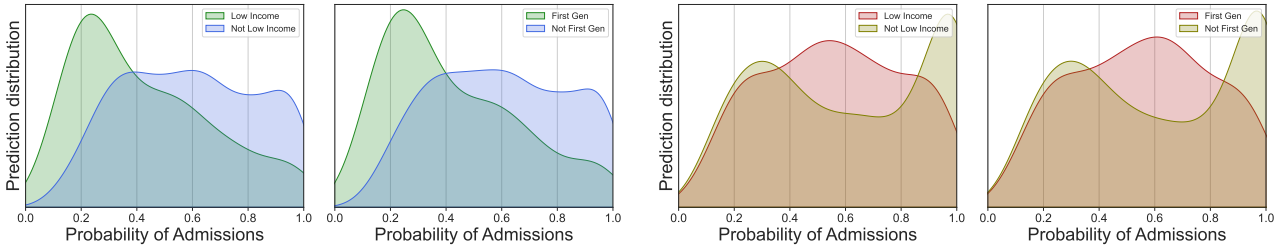
A. Simulation Set-up

Addressing one of the significant challenges in undergraduate student admissions, we explore the presence of socioeconomic bias in the data. For our case study, after a systematic analysis of the input features we consider two binary-sensitive variables namely low-income status and first generation flag. Both the features correlate with reduced access to educational resources and opportunities. In this section, we further discuss and justify our choice of sensitive attributes, providing an in-depth analysis of the results obtained to support our decision. The code and resources for this study are available at our GitHub repository: Fair Evaluator.

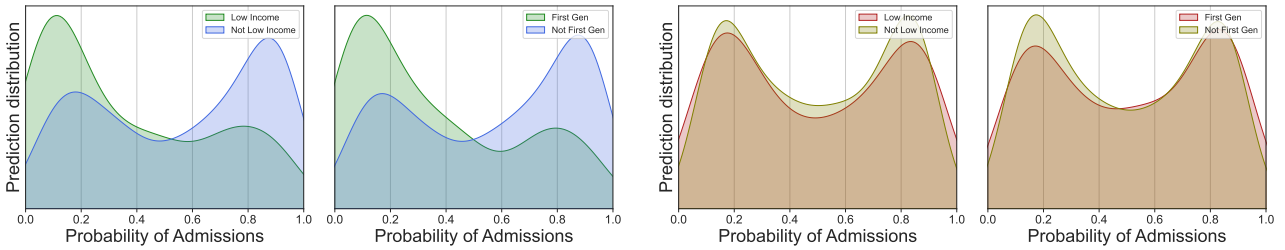
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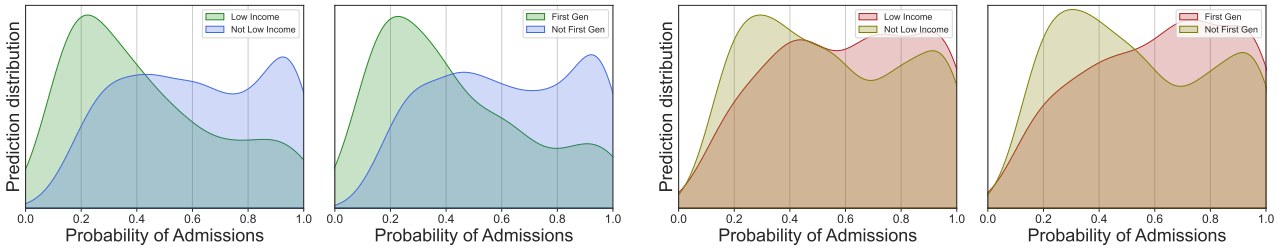
(a) FFNN with Adam optimizer



(b) ICNN with Adam optimizer



(c) FFNN with OAdam optimizer



(d) ICNN with OAdam optimizer

Figure 2: The Kernel Density Estimate (KDE) plots for Feed-forward Neural Network and Input Convex Neural Network. For the two sensitive attributes, namely Low-income Status and First Generation Flag, we showcase both the pre-training and post-training phases of the learning model. Here, the left two columns represent the pre-training phase and the right two columns represent the post-training phase respectively.

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Table III: Fairness Metrics for Deep Learning Models. Columns I and II signify the two sensitive attributes, namely Low-Income Status and First Generation Flag respectively.

Learning Model	*p% Rule*		*Demographic Parity*	
	I	***II***	***I***	***II***
On Pre-Trained Model				
Feed-forward Neural Network	58.236	51.746	0.185	0.196
Input Convex Neural Network	59.01	56.67	0.1646	0.1744
On Fully Trained Model				
Feed-forward Neural Network	98.323	99.617	0.0094	0.0105
Input Convex Neural Network	88.32	87.58	0.1129	0.1147

(a) Using ADAM Optimizer

Learning Model	*p% Rule*		*Demographic Parity*	
	I	***II***	***I***	***II***
On Pre-Trained Model				
Feed-forward Neural Network	56.402	49.304	0.1928	0.2074
Input Convex Neural Network	56.737	55.337	0.1754	0.1810
On Fully Trained Model				
Feed-forward Neural Network	97.69	92.26	0.0050	0.0237
Input Convex Neural Network	96.52	95.35	0.0342	0.0249

(b) Using Optimistic ADAM Optimizer

Given the fairness constraint, we try to achieve a balanced predictive performance by taking both the fairness and performance metrics into account. To evaluate the fairness of our DL models, we utilize two fairness metrics namely *p-%* rule and demographic parity, as discussed in Section IV-C. High *p-%* rule scores are desirable as they indicate minimal disparity in selection rates, signifying a fairer model. Demographic parity, as implemented for our study, calculates the absolute difference between mean predicted outcomes for different groups of the sensitive attribute. So lower demographic parity score indicates reduced disparity between groups, signifying a fairer model. In addition to these, we assess the models using a wide range of performance metrics [5], including accuracy, precision, recall, f1-score and AU-ROC value, to provide a thorough assessment. Among these performance metrics considered, we particularly emphasize the choice of *Recall score* due to its significance in addressing the fairness-accuracy trade-off inherent in the decision-making process [31]. Recall score, also known as sensitivity or true positive rate, measures the ability of a model to correctly identify all the positive instances in the dataset. In a similar context, [20] demonstrates how optimizing for fairness, particularly regarding equal opportunity, might impact the accuracy of the model thereby addressing the fairness-accuracy trade-off. For our case study, the Recall score provides valuable insights into the model's ability to correctly identify positive instances. It ensures that the model maximizes the identification of qualified applicants. This in return mitigates the risk of disproportionately excluding individuals from underrepresented socioeconomic backgrounds.

To evaluate the effectiveness of our debiasing approach, we begin by examining both the initial and final stages of model training. This involves comparing the pre-trained and fully-trained models on the admissions dataset. The *pre-trained models* represent an initial stage in our training process, where the classifier and the adversary are independently trained. The classifier is pre-trained to predict the target variable, while the adversary is pre-trained to predict sensitive attributes from the classifier's outputs. This pre-training serves as a baseline, allowing us to observe the inherent biases before applying the adversarial debiasing technique. Following this, the *fully-trained models* are developed through a comprehensive adversarial training process. During this phase, the classifier is optimized to not only maintain predictive performance but also minimize the adversary's ability to predict sensitive attributes, thereby reducing bias in the predictions. We analyze the fairness metrics, in Table IIIa and Table IIIb, and Kernel Density Estimation (KDE) plots on the pre-trained and fully-trained models, in Figure 2, to affirm the presence of bias in the sensitive features. The comparisons between the pre-trained and fully-trained models, as depicted in the KDE plots in Figure 2, highlights the effectiveness of the debiasing algorithm in mitigating bias over a range of various DNN frameworks. Provided KDE plots are an intrinsic part of the study, effective debiasing is demonstrated by increased overlap or similar distribution between the curves, for each sensitive attribute, indicating that these features have reduced influence over the model predictions. Moreover, we simultaneously try to maintain the predictive performance of the fairness-aware DL models, as seen in Table II, and compare it to that of the unfair baseline models, Table I. By explicitly accounting for these factors in our DL model, we aim to mitigate systematic inequalities and ensure that qualified applicants receive equitable consideration during the decision-making process.

B. Baselines

For the baseline, we have employed several standard models as benchmarks for comparison with our proposed model. The baselines used are as follows,

- Logistic Regression is a classical ML model that is comparatively simple and interpretable. For our study, we employ the model in its standard, unfair configuration [19] to facilitate comparison with more complex models.
- Extreme gradient boost, otherwise known as XGBoost, is a gradient boosting algorithm that is robust and has high predictive performance. Given its ability to handle complex datasets with nonlinear relationships [32], we employ the model's standard, unfair configuration for our study.
- FFNN is a standard architecture in DL capable of modeling complex relationships in datasets while being scalable. For our study, we utilize its unfair version [5]

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as a baseline for getting better insights into our proposed fair model.
- ICNN is a convexity-imposing model that ensures superior representation. Utilizing its unfair version as a baseline for our study [5], allows us to gather insights into the usage of convex optimization techniques in addressing bias.

C. Simulation Results and Discussion

In this section, we provide a performance evaluation of our fairness-aware DL models, focusing on key performance and fairness metrics, and discuss their implications for fairness in the context of student admissions.

We implement the adversarial debiasing technique on the two types of DNN architectures for the study, namely FFNN and ICNN. The models are trained using both Adam and OAdam optimizers separately. In the study, we systematically conduct a performance analysis on both the DNN architectures, implemented with the bias mitigation technique, and showcase their individual performance on the two different types of optimizers. Our analysis presents, as seen in Table I and Table II, the fair ICNN model using the Adam optimizer achieves high recall scores in comparison to the remaining fairness-aware models as well as the baseline models. However, this model is unable to obtain high fairness scores as seen in Table IIIa, and has comparatively lower precision scores. On the other hand, the FFNNs, both using Adam and OAdam optimizers, exhibit high fairness scores while compromising on the recall scores, making them unsuitable. On further investigation, we observe the fair ICNN model using the OAdam optimizer, which, while exhibiting a slightly lower recall score as compared to the fair ICNN model using the Adam optimizer, still outperforms the remaining fairness-aware models as well as the baseline models. Along with that, the fair ICNN model using the OAdam optimizer even performs better in the fairness metric scores, Table IIIb, as compared to the one using the Adam optimizer, Table IIIa. In the fairness context, even though the FFNNs exhibit better performance they have comparatively lower performance metric scores, including the recall scores. While the AU-ROC scores for the ICNN model with OAdam optimizer is lower compared to that of the FFNNs, this performance dip can be attributed to the increased complexity of the ICNN model [26]. As this complexity requires more data to optimize effectively, with an increased dataset it is anticipated that the performance of the ICNN model could be significantly improved, potentially surpassing that of the FFNN [33].

Using the KDE plots, we try to visualize the distributional impact of each sensitive attribute on the learning model. As seen in Figure 2(a) and 2(c), both the fully-trained FFNN models exhibit a higher degree of overlap between the curves for both the sensitive attributes, indicating a reduction in bias and improved fairness. In contrast, the fully-trained ICNN model with the Adam optimizer in Figure 2(b) shows a comparatively lower degree of overlapping. As opposed to its pre-trained model it does not improve the degree of overlap, thus highlighting less effective bias mitigation and potentially less fair predictions. This is an affirmation of the results obtained in Table IIIa. As opposed to the previous models, the fully-trained ICNN model with OAdam optimizer in Figure 2(d) displays a considerable increase in overlapping for the two sensitive attributes when compared to its pre-trained model. This showcases the model to perform successful bias mitigation with respect to the sensitive attributes, which in return is also reflected in its fairness metrics in Table IIIb. The balanced performance of the fair ICNN model using the OAdam optimizer is attributed to the optimizer usage which effectively balances the performance metrics while maintaining high fairness metrics and achieving decent accuracy scores for the study.

In contrast to the baseline models, which are the standard unfair models, our proposed fairness-aware ICNN model optimized with OAdam exhibits a notably high recall score and balanced predictive performance, underscoring the effectiveness of our approach in addressing bias. Moreover, our proposed model demonstrates superior performance compared to the remaining fairness-aware models while trying to strike a balance between fairness and predictive performance. Specifically, it achieves high fairness scores while simultaneously preserving accuracy and maintaining high recall scores, thereby showcasing its ability to achieve both predictive performance and fairness objectives simultaneously.

VI. Conclusion and Future Works

Our study underscores the importance of fairness considerations in the algorithmic decision-making process, particularly in the sensitive domain of student admissions. Given the profound impact undergraduate student admissions have on an applicant's life and future opportunities, ensuring fairness and equity in this process is of utmost importance. Our research addresses this sensitive issue by tackling the challenge of mitigating bias while accounting for the socioeconomically disadvantageous factors present in the data. In our study, we propose a framework, adversarial debiasing technique coupled with OAdam to address the bias present in real-world student admissions data. By prioritizing the recall score as the critical performance metric and leveraging advanced optimization techniques in our proposed DL model, we have tried to address the inherent fairness-accuracy trade-off. Concurrently we also enhance the fairness metrics, the p-% rule score, by an average of 39.989% with respect to the sensitive attributes. In summation, our research focuses on improving the student admissions decision-making process while ensuring a fair and equitable system.

For future work, we plan to explore the integration of causal inference techniques into fairness-aware ML models. Additionally, we seek to develop causal frameworks that could provide deeper insights into fairness beyond correlation-based approaches in the context of student admissions. Furthermore, we acknowledge the importance of validating our method on multiple datasets. Given the real-world relevance and richness of features in our current dataset, future research will focus

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on acquiring and preprocessing additional datasets to ensure the broader applicability and robustness of our results.

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